

Report on House Price Prediction

1. Introduction

The project aimed to create a machine learning model to predict house prices based on factors like size, number of bedrooms, age, and downtown proximity, involving data exploration, preprocessing, and evaluation.

2. Data Exploration and Visualization

Steps Taken:

- Loaded the dataset and explored its structure using `head()` and `describe()` methods.
- Identified and handled missing data by replacing them with the column means.
- Removed outliers using the IQR method to ensure robust model training.
- Visualized relationships between features and the target variable (`Price`) using scatter plots.

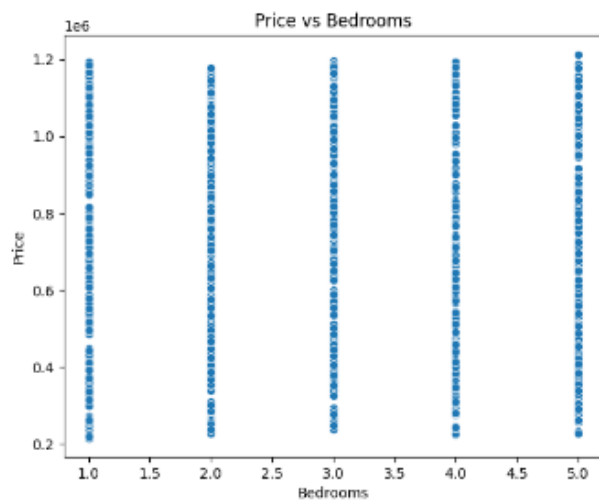
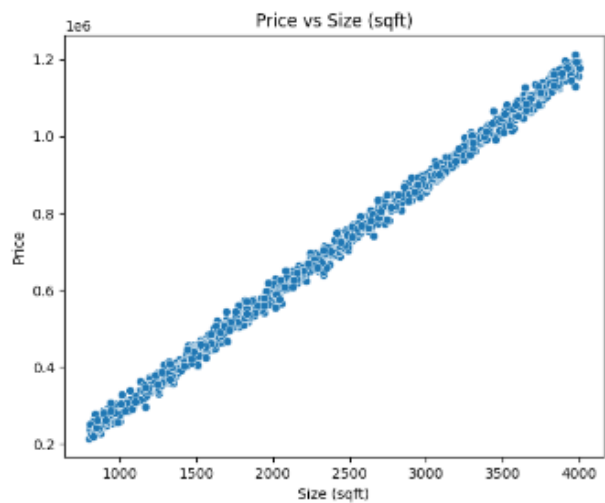
Key Insights:

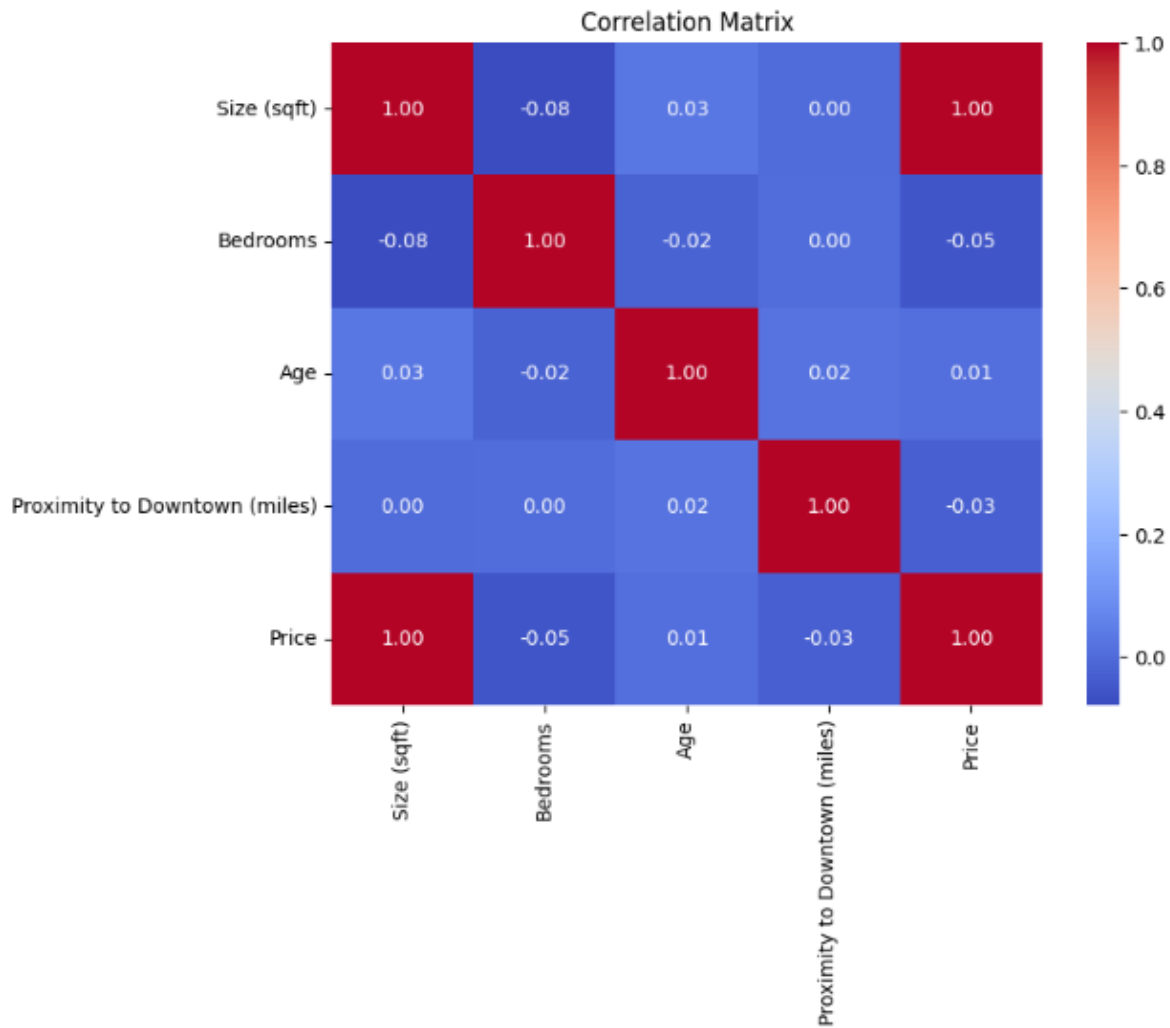
- **Positive Correlations:** Larger house size and more bedrooms showed a positive relationship with price.
- **Negative Correlations:** Older homes and greater distance from downtown were negatively correlated with price.
- Correlation matrix provided a holistic view of relationships between variables, aiding feature selection.

Challenges:

- Identifying and visualizing outliers required careful handling to balance data quality without losing meaningful variations.

Plots:





3. Data Preprocessing

Steps Taken:

- Split the data into training and testing sets (70% training, 30% testing).
- Standardized the features to ensure uniform scaling using `StandardScaler`.

Challenges:

- Ensuring proper scaling without data leakage (applied scaling parameters from training data to test data).
 - Balancing the dataset split to avoid overfitting or underfitting.
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4. Model Development

Linear Regression:

- Trained a linear regression model using standardized training data.
- Used `statsmodels` to assess the statistical significance of coefficients.
- Extracted model coefficients and intercept for interpretation.

Challenges:

- Multicollinearity among features could impact coefficient estimates, though the small number of features mitigated this issue.
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5. Model Evaluation

Metrics:

- **Mean Squared Error (MSE)**
- **R-squared**
- **Adjusted R-squared:** Adjusted for the number of predictors and data points.

Visualizations:

- A scatter plot of predicted vs. actual prices highlighted the model's performance, with a line of perfect prediction providing a benchmark.

Insights:

- The model showed reasonable predictive accuracy, with R^2 indicating the proportion of variance explained by the features.
- Residual analysis revealed areas where the model could be improved, such as incorporating additional features.

Plot:



6. Model Improvement Attempts

- **Outlier Handling:** Improved model robustness by removing extreme values using IQR.
- **Feature Engineering:** Future iterations could include categorical data like neighborhood quality or additional numerical features like lot size.
- **Regularization:** Incorporating Lasso or Ridge regression to handle potential multicollinearity.

7. Conclusion

Applicability:

- The model is suitable for estimating house prices in the given dataset but may not generalize well to all markets without retraining on localized data.

Limitations:

- Lack of categorical data or external economic factors limits the scope of predictions.
- Simplistic feature selection may omit important determinants of house prices.

Future Work:

- Incorporate external data sources, such as economic indicators or market trends.
- Experiment with more complex models like decision trees or ensemble methods (e.g., Random Forests).